



DANIEL MOON

Undergraduate Student

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EDUCATION

Calcutta International School

Valedictorian
2007-2019

Seoul National University Electrical and Computer Engineering

Mar 2020~Present
(1 semester until graduation)

RESEARCH INTERESTS

Computer Architecture

Computer System

SoC Design

FPGA-Based AI Accelerators

Diffusion-based Language Models

ABOUT ME

Passionate and detail-oriented senior at Seoul National University, specializing in computer hardware with a focus on AI accelerators, particularly FPGA and ASIC development.

Having been born and raised in India and later moving to South Korea for university, I have developed a global perspective and the adaptability to grow in a new academic and cultural environment. Alongside my technical interests, I am keenly interested in market and company research, aiming to blend my hardware expertise with insights into industry trends and business strategies.

Eager to gain hands-on experience in hardware-focused research labs and companies, I am particularly interested in deepening my knowledge in AI-specific hardware-software co-design, neural processing units, and optimization techniques for low-power, high-efficiency computing. Additionally, I am keen to expand my understanding of AI software, such as neural network architectures, machine learning frameworks, and their applications in embedded systems.

Through coursework and hands-on engineering experience, I have built a solid understanding of digital circuits, microarchitecture, kernel design, and resource allocation. This background has trained me to approach hardware problems with a system-level perspective, identify performance bottlenecks, and design under strict resource constraints. As a result, I am able to connect AI model characteristics to practical decisions in NPU/SoC architecture, resource scheduling, and performance optimization, which makes me a strong fit for AI hardware and digital system design roles.

WORK EXPERIENCE

Computer Architecture and Parallel Processing Lab, CAPP Lab
Seoul National University
Research Intern

July 2024 – Sept 2024
Prof. advisor:
Taeho Lee,
Nguyen Truong

- Host Code Optimization for Real-Time Object Detection (YOLOV3Tiny)
- Programming and Testbench Verification for 3D Unet NPU Conv Module
 - Implemented on AMD/Xilinx Zynq+ ZCU106 FPGA Board

HyperAccel
South Korean NPU Startup
System Team Intern

Dec 2024 – July 2025

- Runtime host code programming and .ila debugging
 - LPU architecture migration from :
 - AMD Alveo U55C to AMD Alveo V80
 - AMD Alveo U55C to Achronix Speedster7t
 - AMD Alveo V80 to AWS EC2 F2 instances

Compute Memory Architecture Lab, CMA Lab
Seoul National University
Research Intern

July 2025 – Dec 2025
Prof. advisor:
Sungjoo Yoo

- Efficient MXFP Inference Framework
- Diffusion LLM Inference Optimization Schemes

Very Large Scale Integration Lab, VLSI Lab
Seoul National University
Research Intern

July 2025 – Dec 2025
Prof. advisor:
Jaejoon Kim

- Diffusion LLM Quantization Framework
 - Outlier-Aware Mixed Precision Quantization
 - Early-Exit Convergence
 - Activation-Guided Quantization Calibration

PUBLICATIONS

An Energy-Efficient Daily Surveillance System with DVS-CIS Sensor Fusion and Event-based NPU Triggering

Fall 2024

Accepted for ISCAS 2025 (Lecture Session and Poster Presentation)
3rd Author

PROJECTS AND RESEARCH

- Self Improving AI Competition** | *ETRI Fashion-How* Summer 2024
Image Classification and Chatbot for Low Capacity Embedded Systems
- CIS-DVS Sensor Fusion System** | *CAPP Lab* Fall 2024
Efficient Image Recognition with a CIS-DVS Heterogenous Sensor Fusion System
- CNN Inference Architecure Design** | *AIX Design Contest* Summer 2025
End-to-End CNN Architecture Design on Nexys-A7-100T FPGA Board
- Social Media-Driven Financial Market Prediction** | *Seoul National University* Spring 2026
Developed the backend infrastructure and core prediction algorithms for a stock market forecasting platform.

EXTRA CURRICULAR ACTIVITIES

SURI (Student Union for Reasonable Investment)

Strategy Department

- Market Research
- Company Analysis Report
- Social Event Hosting

Samsung Shining Star

Team Captain

- Semi-Conductor integrated Product Development
- Final Standing: 3rd Place

MILITARY SERVICE

Republic of Korea Army

Sept 2021 - Mar 2023

Sergeant

- Ministry of National Defense Honor Guard

SKILLS

Languages : English (Native), Korean (Native), Hindi (3rd Language)

Programming : C/C++, Python, Verilog

Tools : Git, Vivado, LaTeX